

Markscheme

Mathematics: analysis and approaches
Practice question bank

66. (a)

The 3rd term of an arithmetic sequence is **17** and the 9th term is **53**.

Find the sum of the first 20 terms.

$$u_9 - u_3 = 6d: 53 - 17 = 6d \Rightarrow d = 6 \quad (M1) \text{ A1}$$

$$u_1 = u_3 - 2d = 17 - 12 = 5 \quad (M1)$$

$$S_{20} = \frac{20}{2} (2(5) + 19(6)) \quad (M1)$$

$$= 1240 \quad \text{A1}$$

[5 marks]